

Gokul Krishna R

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EDUCATION

Cochin University of Science and Technology
MSc Computer Science (with Specialization in Data Science)
Sacred Heart College
BSc Computer Applications (Triple Main)

Kochi, Kerala
Jun. 2025 – April 2027
Kochi, Kerala
Jul. 2022 – May 2025

EXPERIENCE

IBM SkillsBuild Virtual Internship
Data Analytics Intern

Dec. 2024
Remote

- Gained insights about various data analysis and visualization techniques used in the industry
- Worked with hands-on masterclasses and practical sessions
- Improved understanding of data interpretation and report submission workflows

Klickevents Infosolutions
Python Intern

May 2024
Kochi, Kerala

- Worked with the pandas library in Python for data analysis and visualization
- Performed data preprocessing and exploratory data analysis tasks
- Developed practical understanding of Python-based data workflows

PROJECTS

Federated Learning and Machine Unlearning Framework | *Python, PyTorch, Federated Learning, Deep Learning*

- Developed a Federated Learning framework with non-IID client data distribution using PyTorch
- Implemented machine unlearning techniques including Gradient Ascent (GA), Negative Gradient Plus (NG+), Loss-based NG+, and Fisher Forgetting (FF)
- Evaluated forgetting effectiveness using global accuracy, client-level accuracy, confusion matrices, and performance analysis
- Conducted experiments on MNIST and CIFAR datasets to study privacy-preserving learning and selective data removal

Auto Traffic (Traffic Sign Recognizer) | *Python, TensorFlow, CNN, Django*

- Developed a Traffic Sign Recognition system using TensorFlow and CNN
- Performed image preprocessing including resizing, normalization, and augmentation
- Trained the model on the GTSRB dataset for traffic sign classification
- Deployed the solution using a Django web application

AI Resume and Portfolio Builder | *Python, Gradio, Hugging Face, Transformers*

- Developed a hybrid AI web application to generate ATS-friendly resumes, cover letters, and portfolio content
- Built portfolio content generation features tailored for students and freshers
- Integrated Transformer-based LLMs through the Hugging Face Inference API
- Implemented a template-based fallback system and deployed the application on Hugging Face Spaces

CarbonAnalyzer: Personal Carbon Footprint Analyzer | *Next.js, TypeScript, Tailwind CSS, Hugging Face, Recharts*

- Developed an AI-powered web application to calculate and analyze personal carbon emissions across transportation, energy, food, consumption, waste, and lifestyle
- Implemented interactive dashboards with real-time charts, Eco Score, carbon budget tracking, and a What-If Simulator for sustainability planning
- Integrated Hugging Face Inference API to generate personalized carbon reduction recommendations with a rule-based fallback engine
- Built privacy-first features including client-side calculations, PDF report generation, and local data persistence without requiring a database

TECHNICAL SKILLS

Languages: Python, C, C++, SQL, HTML/CSS

Libraries & Frameworks: Pandas, TensorFlow, CNN, Django, Gradio, Hugging Face Transformers

Core Skills: Data Visualization, Machine Learning, Deep Learning, NLP, LLMs, Prompt Engineering, Feature Engineering, Data Preprocessing

Tools: MySQL, Microsoft Word, Microsoft PowerPoint, Microsoft Excel